

CompTIA.

A+ Core 1

Exam 220-1201

A+ 220 1201

Module 9



Supporting Mobile Devices

Learning Outcomes

Prepare for A+ Core 1 by:

- Setting up mobile devices and peripherals.
- Configuring mobile device apps.
- Installing and configure laptop hardware.
- Troubleshooting mobile device issues.

Lesson 9.1

Mobile Devices and Peripherals

Learning Outcomes

What is the little nub that is embedded into the keyboard on some laptops that controls the mouse cursor called?

What option on a mobile device disables some, or all, of the wireless features (cellular data, Wi-Fi, GPS, Bluetooth, and NFC), depending on the device type and model?

What is the small card that is used to identify the mobile device on the mobile network called?



Learning Outcomes



You are in a remote location and want to use your mobile device's cellular connection on your laptop wirelessly. What option do you need to enable and configure on the mobile device?



When you make a purchase using contactless payment on your cell phone, what technology is being used?

Mobile Devices and Peripherals



MOBILE DEVICE
ACCESSORIES



MOBILE DEVICE
WIRED CONNECTION
METHODS



PORT REPLICATORS
AND DOCKING
STATIONS



WI-FI NETWORKING

Mobile Devices and Peripherals



Cellular Data
Networking



Mobile Hotspots and
Tethering



Bluetooth Wireless
Connections



Near-Field
Communication
Wireless Connections



Input devices

Touchpad
Trackpad
Drawing pad
Track point



Stylus

Touchpen used
on drawing
pads and
touchscreen



A/V input and output

Microphone,
speakers,
digital camera,
web
conferencing

Mobile Device Accessories

Mobile Device Wired Connection Methods

Laptop Ports

- Video port, USB Type A ports, USB Type C ports, microphone and speaker jacks, RJ45, memory card reader

Smartphone and Tablet Connectors

- USB-C
- Micro-B USB and Mini-B connector

Port Replicators and Docking Stations

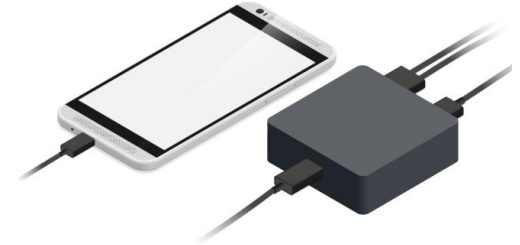
A port replicator



A laptop docking station



Example of a smartphone dock



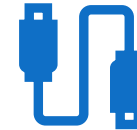
Port replicator

Provides full complement of ports



Docking station

Port replicator, can support laptop to operate like desktop



Smartphone and tablet docks

Connects device to monitor, external speakers, keyboard/mouse input devices

Wi-Fi Networking



Enabling and disabling Wi-Fi



Airplane mode

Disables some or
all of Wi-Fi



Wi-Fi antenna
connector/placement

On mobile
devices, two
wires connected
to Wi-Fi chip,
allows device to
get strongest
signal

Cellular Data Networking

Cellular networking technologies

3G: used in first iPhones and smartphones, up to 7.2 Mbps

4G: has LTE technology, supports speeds to 300 Mbps

5G: offers download speeds to 10 Gbps

SIM/eSIM

Card allows device to connect and access functions

Enabling and disabling cellular data



Configuring tethering on an Android phone



Mobile Hotspots and Tethering

Tethering

Connecting another device such as a laptop to a mobile device to share its cellular data connection

Can use USB cable, Bluetooth, or Wi-Fi

Hotspot

Tethering over a Wi-Fi connection

Bluetooth Wireless Connections

Bluetooth radio creates short-range PAN to share data

Enabling Bluetooth

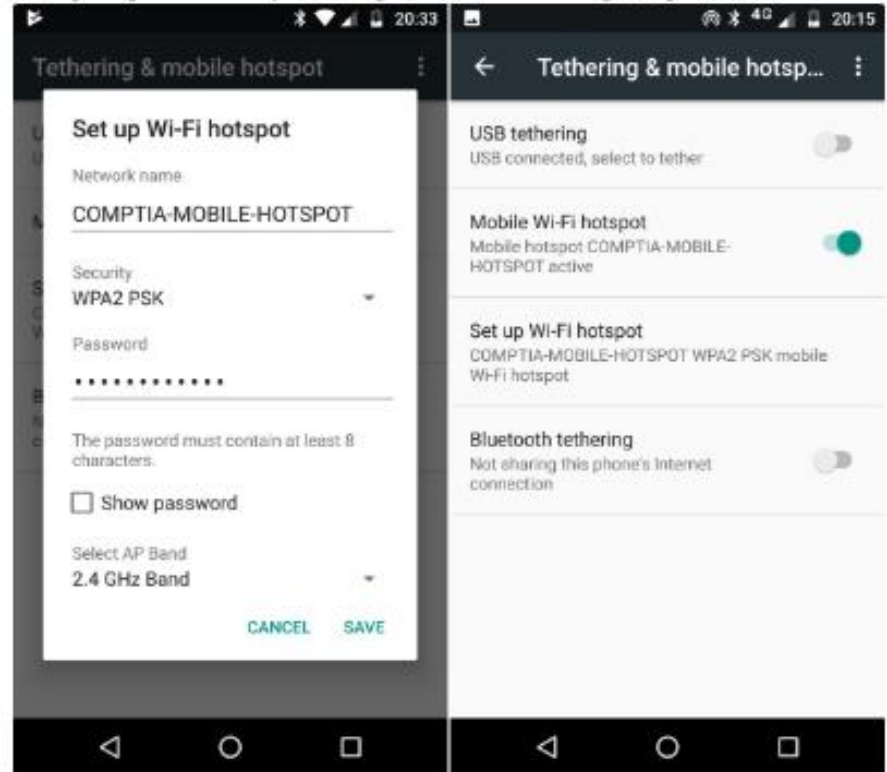
Enable pairing

In iOS: via Settings to General to Bluetooth

In Android: via the notification shade

In Windows: via the applet in Control Panel

Configuring mobile hotspot settings (left), then enabling it (right)



NFC mobile payment



Near-Field Communication Wireless Connections

Functions

Allows for very short-range transmission to activate the reader chip in contactless reader

Allows mobile device to make payments via contactless point of sale

Also used to configure other connections such as pairing Bluetooth

Lab Activity

- Lab: Manage Mobile Devices
(available in CertMaster Learn and CertMaster Perform)
- Configure a Laptop Dock and External Peripherals
(available in CertMaster Learn and CertMaster Perform)

Lesson Review: Mobile Devices and Peripherals

- Mobile Device Accessories
- Mobile Device Wired Connection Methods
- Port Replicators and Docking Stations
- Wi-Fi Networking
- Cellular Data Networking
- Mobile Hotspots and Tethering
- Bluetooth Wireless Connections
- Near-Field Communication Wireless Connections

Lesson 9.2

Mobile Apps and Data

Mobile Apps (Slide 1 of 3)

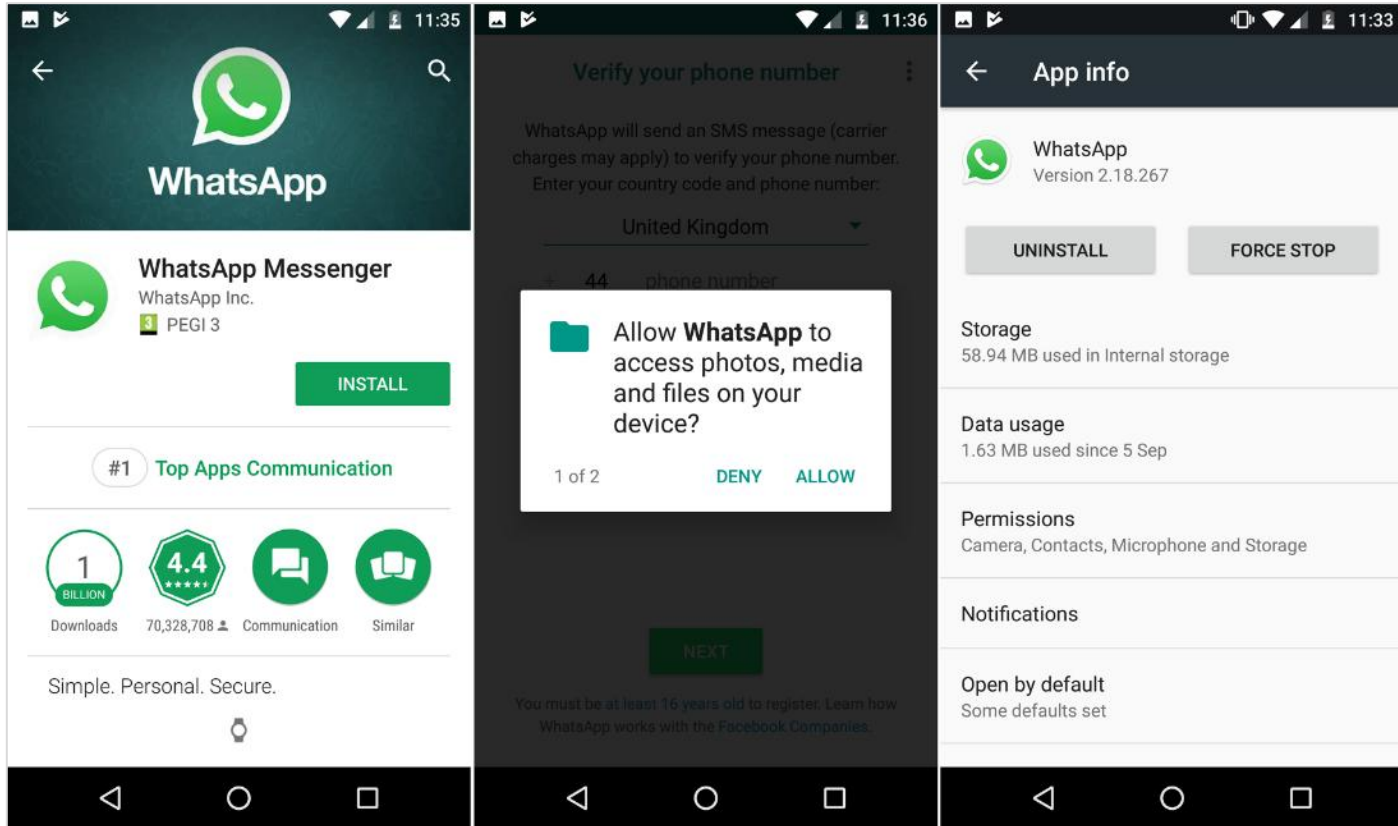
- iOS apps- downloadable from Apple's app store
 - Xcode IDE and Swift language
- Android apps- downloadable from Google Play store and third-party sites
- App permissions- allow access to location, networks, camera, microphone, etc.

Mobile Apps (slide 2 of 3)



Screenshot reprinted with permission from Apple Inc., and WhatsApp.

Mobile Apps (Slide 3 of 3)



Screenshots courtesy of Android platform, a trademark of Google LLC., and WhatsApp.

Account Setup

- Apple ID
- Google
- Synchronize settings and data between multiple devices



Screenshot reprinted with permission from Apple Inc.

Types of Data to Synchronize

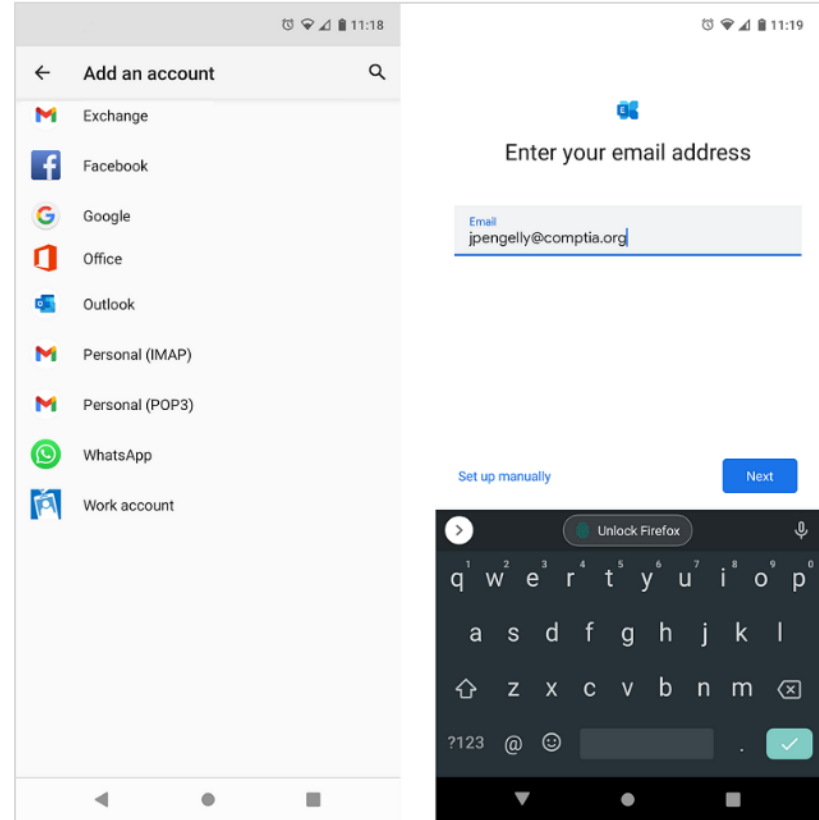
- Contacts
- Calendar
- Mail
- Multimedia
- Documents
- Apps
- Passwords

Email Configuration Options (Slide 1 of 3)

- Commercial email
 - Google, Outlook, Office, Yahoo, etc.
- Enterprise email
 - Microsoft Exchange
 - ActiveSync

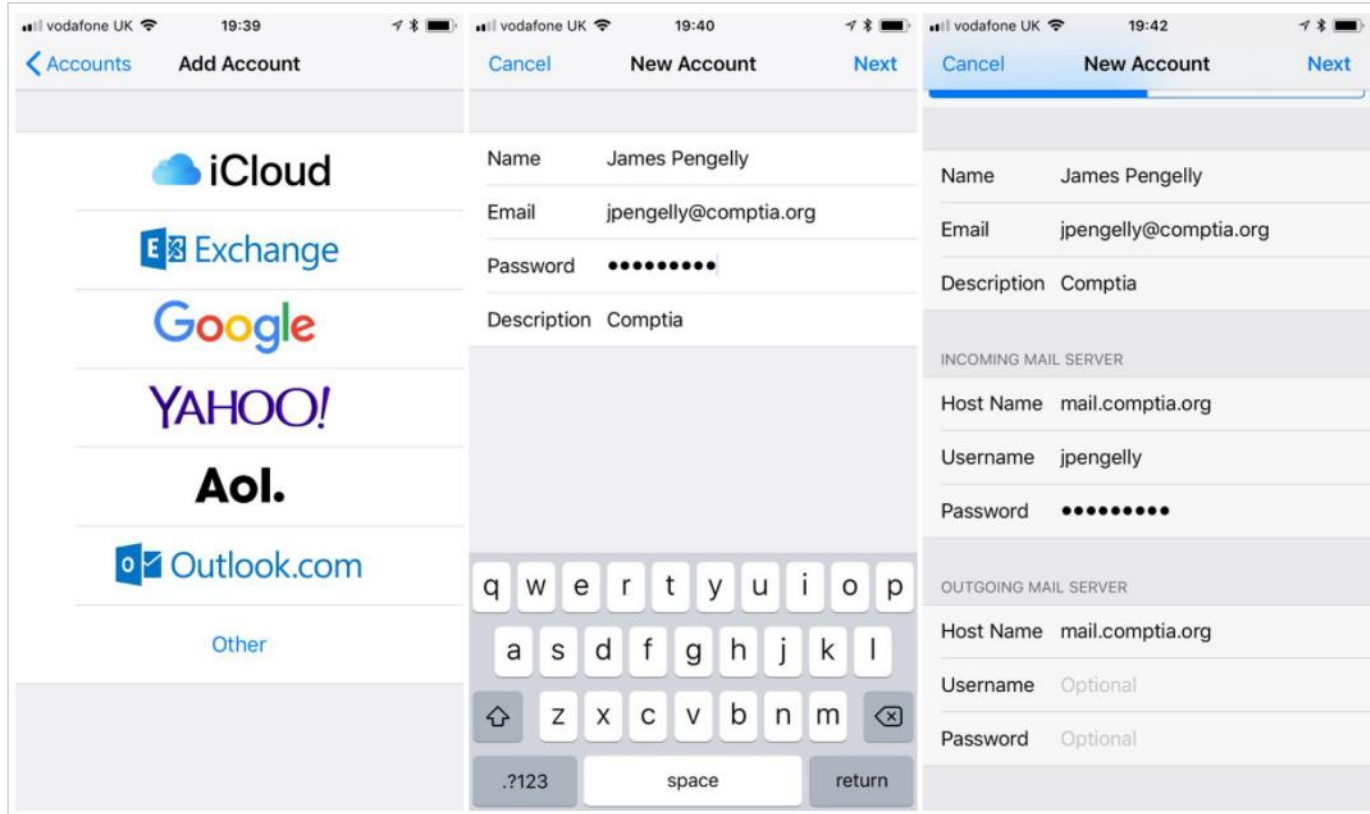
Email Configuration Options (Slide 2 of 3)

- AutoDiscover
- Manual
 - SMTP
 - IMAP
 - POP3



Screenshot courtesy of Android platform, a trademark of Google LLC.

Email Configuration Options (Slide 3 of 3)

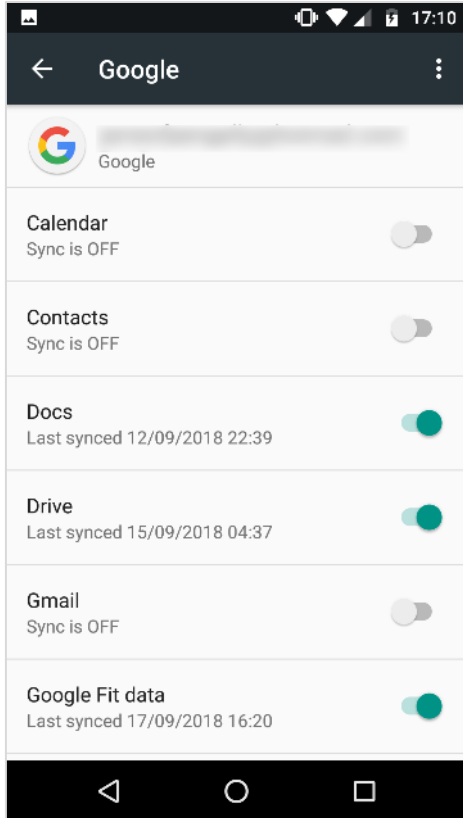


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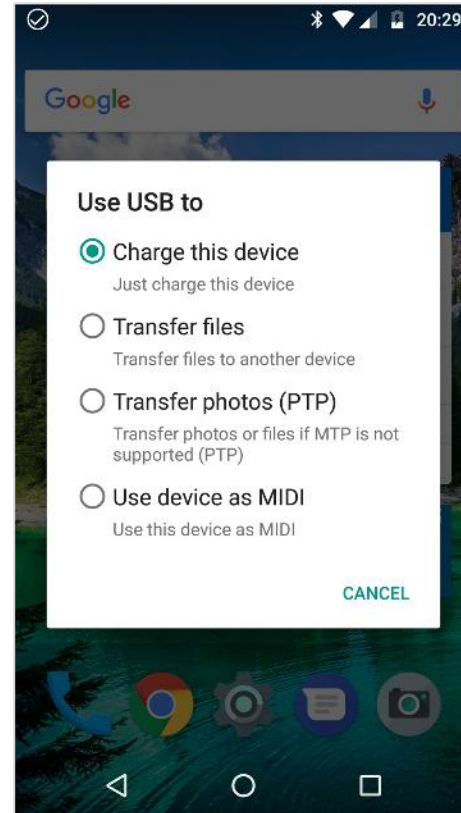
Synchronization Methods (Slide 1 of 2)

- Sync to PC
 - USB or wireless
- Sync to Automobile
 - CarPlay
 - Android Auto

Synchronization Methods (Slide 2 of 2)



Screenshot courtesy of Android platform, a trademark of Google LLC.



Screenshot courtesy of Android platform, a trademark of Google LLC.

Enterprise Mobility Management (Slide 1 of 2)

- EMM- Enterprise Mobility Management
 - MDM- Mobile Device Management
 - Networks, hardware use, reset and remote wipe
 - MAM- Mobile Application Management
 - Application settings and control
- Apple Business Manager
- Managed Google Play

Enterprise Mobility Management (Slide 2 of 2)

The screenshot displays the Microsoft Azure portal interface for configuring Enterprise Mobility Management (EMM) profiles. The breadcrumb navigation at the top indicates the path: « Device configuration - Profiles > Create profile > Device restrictions > Restricted Apps. The interface is divided into three main panels: 'Create profile', 'Device restrictions', and 'Restricted Apps'.

Create profile panel:

- Name:** gtslearning EMM Default Android Policy (with a green checkmark)
- Description:** Enter a description... (with a green checkmark)
- Platform:** Android (selected from a dropdown)
- Profile type:** Device restrictions (selected from a dropdown)
- Settings Configure:** A blue button with a right-pointing arrow.
- Create:** A blue button at the bottom.

Device restrictions panel:

Select a category to configure settings.

- General:** 11 settings available
- Password:** 10 settings available
- Google Play Store:** 1 setting available
- Restricted Apps:** 2 settings available (highlighted in blue)
- Browser:** 5 settings available
- Allow or Block apps:** 3 settings available
- Cloud and Storage:** 4 settings available

Restricted Apps panel:

Use these settings to stay informed about which users install apps that are not approved for use your company. Select the type of restricted app list:

Prohibited apps - A list of apps that you want to be informed about when users install them.
Approved apps - A list of apps that are approved for use in your company. When users install an that is not in this list, you will be informed.

Type of restricted apps list: Not configured

* Apps list (with a help icon):

Import Export

App URL	App Bundle ID	App Name	Publisher	
e.g. https://pla	e.g. com.apple	Not configurec	Not configurec	Add
APP URL	APP BUNDLE ID	APP NAME	PUBLISHER	
No apps				

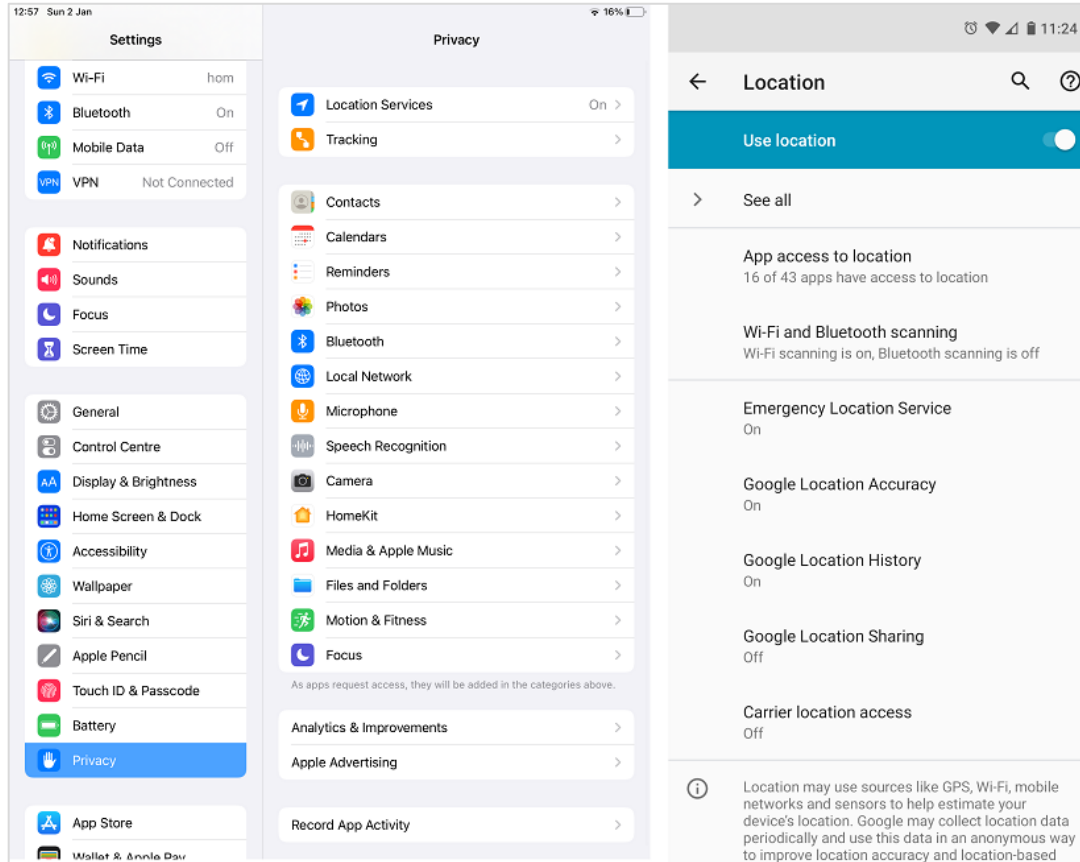
OK: A blue button at the bottom.

Screenshot courtesy of Microsoft.

Location Services (Slide 1 of 2)

- Global positioning system
 - Satellite connectivity to determine location
- Indoor positioning system
 - Cellular towers, Wi-Fi access points, and Bluetooth/RFID beacons

Location Services (Slide 2 of 2)



Screenshots reprinted with permission from Apple Inc., and Android platform, a trademark of Google LLC.

Lesson Review: Mobile Apps and Data

- Mobile Apps
- Account Setup
- Types of Data to Synchronize
- Email Configuration Options
- Synchronization Methods
- Enterprise Mobility Management
- Location Services

Lesson 9.3

Laptop Hardware

Laptop Disassembly Processes (Slide 1 of 2)

- Smaller size components
- Hand tools
- Parts
- Form factors and plastics
 - Easily damaged

Laptop Disassembly Processes (Slide 2 of 2)

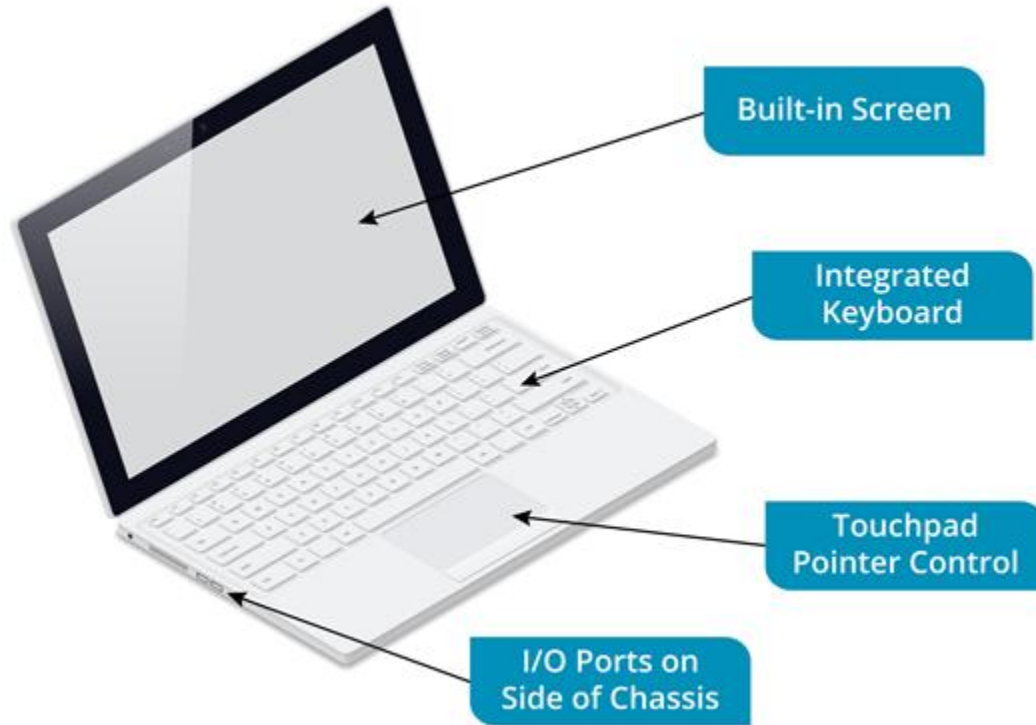


Image © 123RF.com

Battery Replacement (Slide 1 of 3)

- AC Adapter- power supply
- Battery removal
 - Release tab
 - Screws



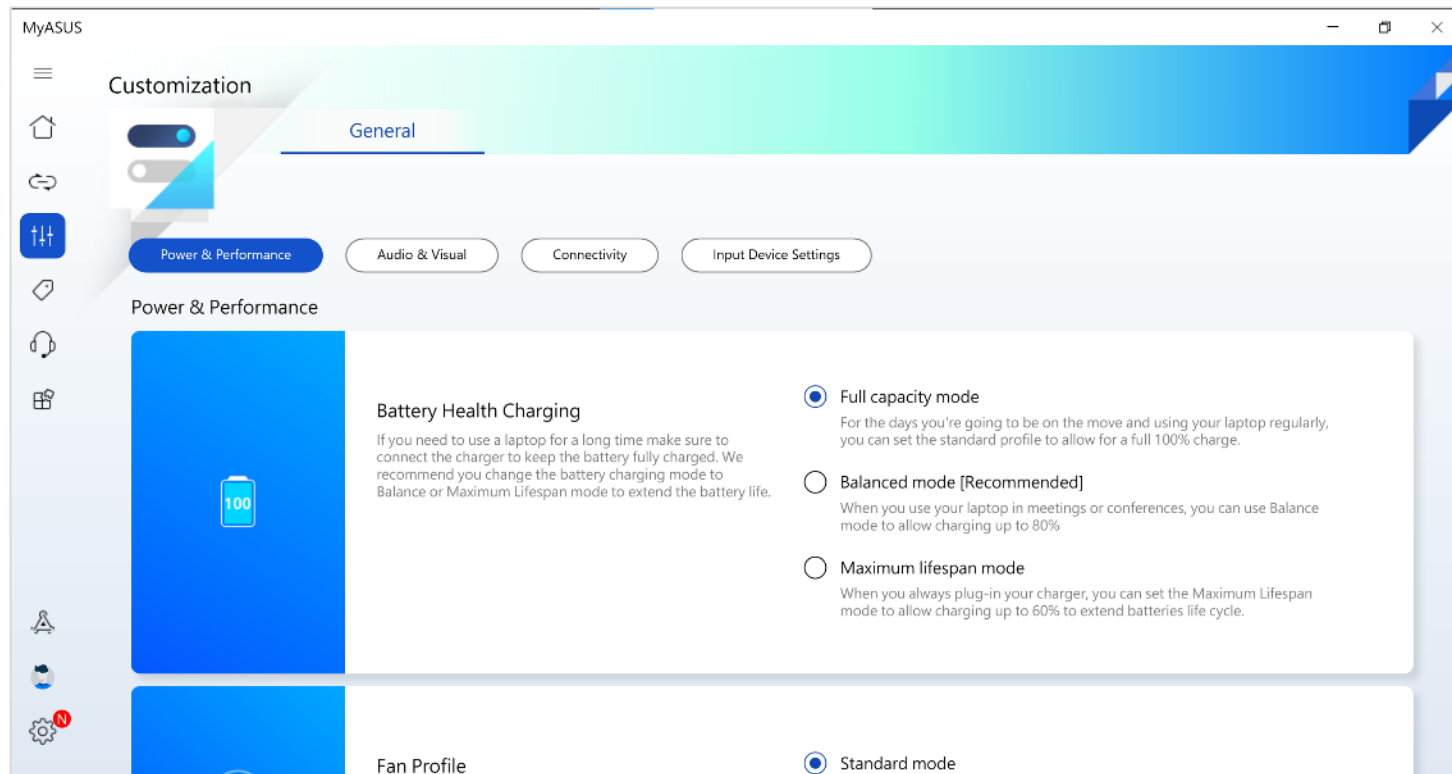
Image by Olga Popova © 123RF.com

Battery Replacement (Slide 2 of 3)



Image by cristi180884 © 123RF.com

Battery Replacement (Slide 3 of 3)



Screenshot used with permission from ASUSTek Computer Inc.

RAM and Adapter Replacement (slide 1 of 2)

- Upgrading RAM
- Limited number of slots
- SODIMM



Image courtesy of CompTIA.

RAM and Adapter Replacement (Slide 2 of 2)



Image courtesy of CompTIA.

- Adapter cards
 - Mini PCIe
 - mSATA
 - M.2
- Wireless

Disk Upgrades and Replacement (Slide 1 of 2)

- Migration of data
- Replacement of drive
- 2.5" HDD
- SSD
 - mSATA
 - M.2

Disk Upgrades and Replacement (Slide 2 of 2)



Image © 123RF.com

Keyboard and Security Component Replacement

- Full disassembly may be required for replacement
- Key replacement for cleaning
- Biometric security component
- Camera and microphone replacement
 - Removal of screen bezel
- NFC scanner

Lesson Review: Laptop Hardware

- Laptop Disassembly Processes
- Battery Replacement
- RAM and Adapter Replacement
- Disk Upgrades and Replacement
- Keyboard and Security Component Replacement

Lesson 9.4

Troubleshoot Mobile Devices

Power and Battery Issues (Slide 1 of 2)

- Check power supply and outlet for proper voltage
- Check battery is correctly inserted
- Battery health condition



Image by gasparij © 123RF.com

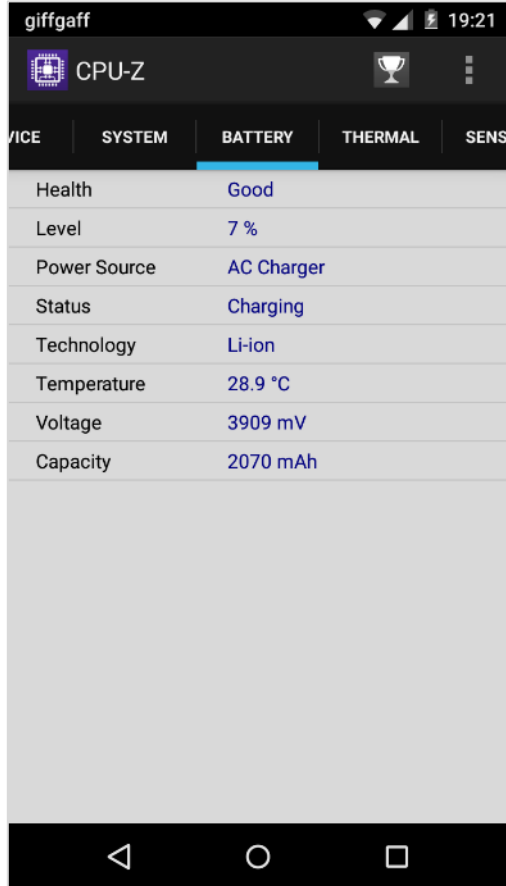
Power and Battery Issues (slide 2 of 2)

- Improper charging
 - Charging when battery is not fully discharged can shorten lifespan
 - Intelligent charge monitoring systems
- Swollen battery
 - Swelling and bulging of a battery is a sign of failure
 - Battery should be removed immediately and disposed of properly



Screenshot reprinted with permission from Apple Inc., and Android platform, a trademark of Google LLC.

Hardware Failure Issues



The screenshot shows the CPU-Z application interface. At the top, the status bar displays 'giffgaff', signal strength, Wi-Fi, and the time '19:21'. The app header includes the CPU-Z logo and a trophy icon. Below the header is a navigation bar with tabs: 'DEVICE', 'SYSTEM', 'BATTERY' (which is highlighted), 'THERMAL', and 'SENSORS'. The main content area displays battery-related information in a table format.

Health	Good
Level	7 %
Power Source	AC Charger
Status	Charging
Technology	Li-ion
Temperature	28.9 °C
Voltage	3909 mV
Capacity	2070 mAh

At the bottom of the screen, the standard Android navigation bar is visible with back, home, and recent apps buttons.

- Overheating
 - Cooling pad
- Liquid damage
 - IP scale of waterproofing
- Physical damage to external ports
 - Dust and moisture exposure

Screen and Calibration Issues

- Backlight failure
- Broken screen
 - Glass can cut and injure, replace at earliest possible opportunity
- Digitizer damage
 - Touch input
 - If damaged, device input will not work
- Cursor drift/touch calibration
 - Driver updates
 - Screen protectors and phone cases can reduce sensitivity

Connectivity Issues

- Interference
- Antenna connectors are firmly seated
- Network connectivity testing



Screenshot used with permission from M2Catalyst, LLC

Malware Issues

- Data collection and external reporting
- Background processes
- Data transfers over limits
- Hardware permission changes
 - Accessing network and hardware resources without proper permissions
- Malware may block new apps downloads

Lab Activity

- Challenge Lab: Mobile Hardware Support
(available in CertMaster Learn and CertMaster Perform)

Lesson Review: Troubleshoot Mobile Devices

- Power and Battery Issues
- Hardware Failure Issues
- Screen and Calibration Issues
- Connectivity Issues
- Malware Issues